

Seat. No. _____

KADI SARVA VISHWAVIDYALAYA
BE SEMESTER-VIII (New) EXAMINATION-APRIL 2025

Subject Name: BIG DATA ANALYTICS

Subject Code: CE802-N

Date: 08/04/2025

Time: 12:30pm – 03:30PM

Total Marks: 70

Instructions:

1. Answer each section in separate answer sheet.
2. Use of scientific calculator is permitted.
3. All questions are Compulsory.
4. Indicate clearly, the option you attempt along with its respective question number.
5. Use the last page of main supplementary of rough work.

Section-I

- Q-1** (A) What is Big data? Explain different types of Big data. [5]
- (B) Difference between Traditional Data vs. Big Data Business Approach [5]
- (C) What is Stream Data? Explain Stream Management System. [5]
- OR**
- (C) Explain the Count-Distinct problem. [5]
- Q-2** (A) Explain the Bloom filter with working in details [5]
- (B) Explain the Flajolet Martin Algorithm. [5]
- OR**
- (A) Explain Structured, Unstructured and Semi Structured Data. [5]
- (B) Explain DGIM Algorithm [5]
- Q-3** (A) Draw and Explain Hadoop Framework in detail. [5]
- (B) Define the term 1. PIG 2. HIVE 3. MONGODB [5]
- OR**
- (A) Discuss about Page Rank Algorithm in detail. [5]
- (B) Define the term 1. Link spam 2. Hubs 3. Authorities [5]

Section-II

- Q-4** (A) Explain Multistage Algorithm in detail. [5]
(B) Explain Multihash Algorithm in detail. [5]
(C) Explain Social Network as Graph [5]

OR

- (C) Explain SimRank in detail. [5]

- Q-5** (A) What is NOSQL? Explain characteristics of NOSQL database. [5]
(B) Define the term 1. Key value Store [5]
2. Graph Store
3. Document Store

OR

- (A) Difference between Master-slave model vs. peer-to-peer model [5]
(B) Explain share nothing Architecture in Big Data. [5]

- Q-6** (A) Explain Large scale file system organization. [5]
(B) Explain the Map Reduce Execution [5]

OR

- (A) Explain Grouping and Aggregation by Map Reduce. [5]
(B) Explain Matrix Multiplication with one Map Reduce Step. [5]

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